Predicting Electronic and Thermodynamic Properties of Functional Materials: A Comparative Study of Graph-Based and Tabular Machine Learning Architectures

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***Abstract*—Traditional Density Functional Theory (DFT) calculations for material discovery are computationally intensive, hindering large-scale screening. This paper proposes a hybrid framework that integrates automated high-throughput DFT workflows with machine learning to predict the properties of perovskites and Heusler alloys. We evaluate and compare graph-based Crystal Graph Convolutional Neural Networks (CGCNN) with tabular ensemble methods, including XGBoost and random forest. Experimental results indicate that CGCNN achieves superior accuracy in predicting electronic properties, such as the Fermi level and Band Gap, by effectively capturing global structural topologies. In contrast, XGBoost demonstrates higher precision for thermodynamic stability (Formation Energy), with feature analysis confirming that localized orbital interactions are the primary physical drivers. This study provides an efficient, dual-model strategy for the automated identification of functional materials, significantly reducing the dependence on expensive quantum mechanical simulations.**

***Index Terms*—High-throughput calculations, density functional theory, machine learning, crystal graph convolutional neural networks, and perovskites.**

I. Introduction

Technological progress in fields such as renewable energy and spintronics relies heavily on the continuous discovery and design of novel functional materials. Two classes of materials, Perovskites and Heusler alloys, have garnered significant attention due to their highly tunable and diverse properties [1], [2]. Perovskites, characterized by the general formula ABX_3, exhibit a vast range of electronic behaviors and tunable bandgaps that are foundational to high-efficiency solar cells [1]. Conversely, Heusler alloys are a cornerstone of spintronics research; their unique half-metallicity allows them to act as conductors for one spin channel and insulators for the other, achieving nearly 100% spin polarization at the Fermi level [2], [3]. However, identifying optimal material candidates within a practically infinite chemical space remains a profound scientific challenge.

Historically, the evaluation of new materials has relied on high-fidelity first-principles calculations, predominantly leveraging Density Functional Theory (DFT) [4]. While DFT provides rigorous quantum mechanical insights into ground-state energies, electronic charge densities, and band structures, it suffers from a critical flaw: computational expense. The time and resources required for structural relaxation and self-consistent field calculations scale poorly with system size, rendering exhaustive, large-scale screening of complex material databases impractical [5], [6].

Recently, machine learning (ML) has emerged as a transformative paradigm to accelerate material discovery [7], [8]. By mathematically approximating complex structure-property relationships, data-driven frameworks can bypass the need for direct, intensive DFT simulations. Despite this progress, a critical research gap persists regarding feature representation [9], [10]. Different physical properties are governed by distinct fundamental interactions. For instance, electronic properties—such as the Fermi level and band gap—are profoundly dictated by the global, three-dimensional geometric topology of the crystal lattice. In contrast, thermodynamic stability, such as formation energy, is primarily driven by localized covalent bonding and valence electron hybridization [11]. Consequently, applying a single algorithmic architecture to predict all material properties often yields sub-optimal results [12].

To address this challenge, this study presents a hybrid predictive framework that couples an automated high-throughput DFT pipeline with a comparative analysis of distinct ML architectures. The primary contributions of this paper are threefold:

1) **Automated Data Generation:** We engineer a robust, multi-stage high-throughput DFT workflow to generate a high-fidelity dataset of electronic and thermodynamic properties for Perovskites and Heusler alloys, minimizing manual calculation inconsistencies [13], [14].
2) **Architectural Comparison:** We systematically evaluate the predictive performance of spatial, graph-based networks—specifically Crystal Graph Convolutional Neural Networks (CGCNN) [15]—against tabular, tree-based ensemble algorithms, including XGBoost [16] and Random Forest [17].

3) **Physical Interpretability:** We demonstrate that while deep learning (CGCNN) excels at mapping global topological traits to electronic properties, ensemble models utilizing engineered Orbital Field Matrices (OFM) offer superior accuracy and physical insights into the localized orbital drivers of thermodynamic stability [11] [18].

II. Methodology

To bridge the gap between traditional quantum mechanical simulations and rapid material screening, this research implements a hybrid methodology. The framework comprises an automated high-throughput Density Functional Theory (DFT) pipeline for ground-truth generation and a dual-representation feature engineering strategy for machine learning (ML) training.

A. High-Throughput DFT Workflow

To generate a high-fidelity dataset for Perovskites and Heusler alloys, we engineered a modular computational pipeline using the Vienna Ab initio Simulation Package (VASP) [4]. The workflow transitions from initial crystal structures to refined electronic properties through four validated stages, as shown on the following and Fig. 1:

1) **Structural Relaxation:** A full geometry optimization is performed to reach the ground-state configuration by relaxing lattice vectors and atomic positions.
2) **Static Self-Consistent Field (SCF) Calculation:** Using the optimized coordinates, a high-precision static calculation determines the converged total energy and electronic charge density [4].
3) **Non-SCF Band Structure Calculation:** Eigenvalues are calculated along high-symmetry k-paths in the Brillouin zone to resolve the energy levels and identify the band gap.
4) **Non-SCF Density of States (DOS):** A dense, uniform k-point mesh is employed to accurately resolve the DOS, providing insights into orbital-specific contributions.

This pipeline is orchestrated via automated Python scripts leveraging the Pymatgen library [13], ensuring numerical convergence and consistency across thousands of structural candidates sourced from the Materials Project [6].

B. Feature Representation and Theoretical Framework

The core of our predictive machine learning framework lies in the divergence of feature representations, tailored to capture different physical interactions:

1) Graph-Based Representation (Spatial Logic): For deep learning via CGCNN, crystal structures are modeled as mathematical graphs. Atoms are defined as nodes and chemical bonds as edges, allowing the model to explicitly capture the 3D periodicity and global topological motifs of the lattice [12], [22]. We hypothesize that this representation is superior for electronic properties (e.g., Band Gap, Fermi Level), which are governed by the long-range spatial arrangement of atoms.

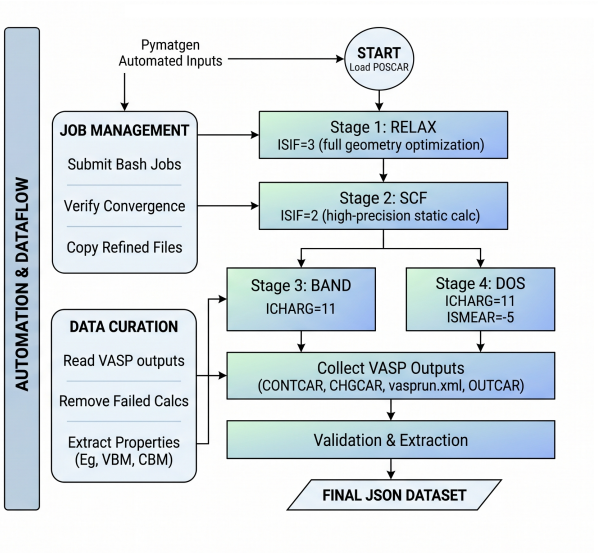


Fig. 1. Detailed Workflow of High Throughput Calculation.

2) Tabular Descriptor Representation (Local Orbital Logic): For tree-based ensemble models like XGBoost [16] and Random Forest [17], structures are encoded into localized physical descriptors. The primary feature is the Orbital Field Matrix (OFM), which mathematically describes the hybridization of outer-shell valence electrons [11]. This is supplemented by Coulomb matrix eigenvalues to capture the electrostatic environment within the lattice [18]. This representation is hypothesized to excel in predicting thermodynamic stability (Formation Energy), as stability is fundamentally rooted in localized covalent and ionic bonding characteristics [10].

III. Experiment Design

To rigorously evaluate the predictive capabilities of the proposed hybrid framework, we designed a comparative machine learning experiment that contrasts spatial graph-based pattern recognition against traditional tabular decision algorithms.

A. Dataset Specification and Curation

Following the automated DFT data generation, custom post-processing scripts were utilized to extract raw data (e.g., `vasprun.xml`, `OUTCAR`) and consolidate it into a unified, machine-readable format. Unconverged or failed calculations were systematically identified and removed to ensure high dataset fidelity. The target variables consisted of continuous electronic properties (Fermi Level, Band Gap, VBM, and CBM) and thermodynamic properties (Formation Energy). To mitigate the risk of overfitting and ensure unbiased evaluation, the curated dataset was partitioned into distinct training, validation, and unseen test sets. The final test sets utilized for comparative evaluation comprised 2,598 structural samples for electronic properties and 2,552 samples for formation energy.

B. Model Implementation

Three distinct machine learning architectures were deployed to predict the target properties. The deep learning

approach, CGCNN, was implemented using the PyTorch library [19] to natively process the 3D crystal graphs. For the tabular-based approaches, the XGBoost [16] and Random Forest [17] ensemble algorithms were trained utilizing the engineered Orbital Field Matrices (OFM) and Coulomb matrix descriptors. Because all target properties are continuous physical quantities, the modeling was framed strictly as a regression task. Mean Squared Error (MSE) was defined as the primary loss function optimized during the training loops. Furthermore, extensive hyperparameter tuning was executed across all three architectures to maximize their respective pattern-recognition capabilities.

C. Evaluation Metrics

The generalization capability of each trained model was assessed on the unseen test set using standard statistical regression metrics. The coefficient of determination (R^2) was employed as the primary metric to quantify the proportion of variance in the physical properties successfully captured by the models. To evaluate the absolute magnitude of prediction deviations, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were calculated. Additionally, a granular analysis of the absolute error distribution—specifically tracking the median absolute error and the 95th percentile error thresholds—was conducted. This distribution analysis is critical for identifying systemic computational limitations and assessing the severity of prediction anomalies at specific energetic boundaries.

IV. Results and Discussion

This section analyzes the predictive performance of the evaluated architectures across distinct electronic and thermodynamic property domains. The results demonstrate that the optimal modeling strategy depends heavily on whether the target property is governed by global structural motifs or localized quantum interactions.

A. Comparative Analysis of Electronic Property Predictions

The electronic properties, including the Fermi level, band gap, Valence Band Maximum (VBM), and Conduction Band Minimum (CBM), were most accurately captured by the graph-based deep learning architecture, as illustrated in Table 1.

- **Fermi Level and Band Gap:** The CGCNN model achieved exceptional predictive fidelity, yielding a coefficient of determination (R^2) of 0.9635 for the Fermi level and 0.9584 for the band gap. In contrast, the tabular XGBoost model recorded significantly lower precision ($R^2 = 0.902$ for Fermi level; $R^2 = 0.797$ for band gap).
- **VBM and CBM:** A similar trend was observed for band edge positions. CGCNN maintained a reliable fit ($R^2 = 0.8564$ for VBM; $R^2 = 0.8744$ for CBM), consistently outperforming Random Forest ($R^2 = 0.781$ for VBM; $R^2 = 0.611$ for CBM).

TABLE I
Predictive Performance Metrics of ML Models for Electronic Properties

Property	Model	R^2	MAE (eV)	RMSE (eV)
Fermi Level	CGCNN	0.9635	0.2211	0.4912
	XGBoost	0.9020	0.3990	0.6900
	Random Forest	0.8710	0.4960	0.7920
Band Gap	CGCNN	0.9584	0.1668	0.3618
	XGBoost	0.7970	0.4970	0.7870
	Random Forest	0.7530	0.5740	0.8680
VBM	CGCNN	0.8564	0.2616	0.6963
	XGBoost	0.8220	0.4790	0.7750
	Random Forest	0.7810	0.5370	0.8590
CBM	CGCNN	0.8744	0.3272	0.8340
	XGBoost	0.7010	0.6530	1.1120
	Random Forest	0.6110	0.7890	1.2690

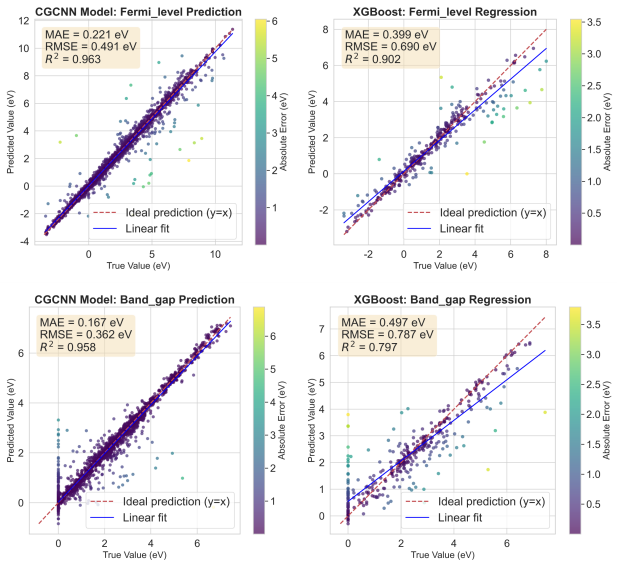


Fig. 2. Regression for the Fermi level and the band gap of CGCNN and XGBoost.

- **Scientific Significance:** The superiority of CGCNN in this domain supports the hypothesis that electronic eigenstates are deeply influenced by the long-range, three-dimensional geometric arrangement of the crystal lattice. However, a persistent vertical distribution anomaly was observed in all models when predicting exact zero-gap states, highlighting a shared computational challenge in resolving the transition boundary between metallic and semi-metallic behaviors using standard DFT outputs.

B. Thermodynamic Stability and the Efficacy of Tabular Models

The prediction of formation energy revealed a significant reversal in model performance. Unlike electronic properties, thermodynamic stability was modeled with much higher precision by tree-based ensemble methods leveraging localized descriptors.

- **Model Performance:** XGBoost emerged as the most accurate architecture for predicting formation energy, achieving an outstanding R^2 of 0.986, followed

closely by Random Forest at 0.973. The graph-based CGCNN, while foundational, showed a drop in performance ($R^2 = 0.9283$) and struggled to quantify extreme thermodynamic stability, as evidenced by prominent outliers in the highly negative energy regime.

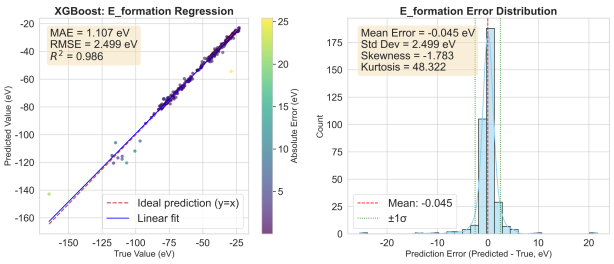


Fig. 3. XGBoost Performance on Energy Formation.

- **Error Distribution:** The error distribution for XGBoost was remarkably tight, with a median absolute error of 0.519 eV, compared to 0.7506 eV for CGCNN.

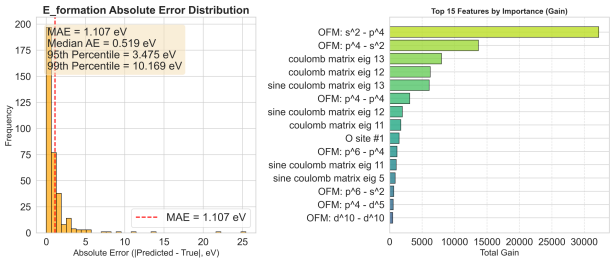


Fig. 4. XGBoost Error and Top Features of Energy Formation.

C. Physical Insights via Feature Interpretability

The high accuracy of tabular models in thermodynamic tasks allowed for a granular analysis of physical drivers through gain-based feature importance.

- **Orbital Dominance:** Across all ensemble evaluations, the importance scores were heavily dominated by Orbital Field Matrix (OFM) descriptors. Specifically, interactions between s and p orbitals (e.g., $OFM : s^2 - p^4, p^4 - s^2$) were identified as the primary determinants of formation energy.
- **Error Distribution:** The error distribution for XGBoost was remarkably tight, with a median absolute error of 0.519 eV, compared to 0.7506 eV for CGCNN.

V. Conclusion and Further Work

This research successfully established an automated, high-throughput hybrid framework that couples Density Functional Theory (DFT) with advanced machine learning architectures to accelerate the discovery of functional materials. By evaluating different data representation strategies, this study reveals critical insights into the structure-property relationships of Perovskites and Heusler alloys, highlighting a phenomenon of "architectural specialization" in materials informatics.

Experimental results demonstrate that the optimal modeling strategy is fundamentally dictated by the underlying physical drivers of the target property. For electronic characteristics, such as the Fermi level, band gap, and band edge positions, the spatial graph-based CGCNN architecture exhibits a significant predictive advantage. This supports the theoretical premise that electronic eigenstates are heavily influenced by the global, three-dimensional geometric topology of the crystal lattice. Conversely, when predicting thermodynamic stability, such as formation energy, traditional tree-based ensemble models (e.g., XGBoost and Random Forest) demonstrate superior performance. Feature importance analyses confirm that thermodynamic stability is primarily governed by localized orbital interactions and valence shell hybridization, which are highly effectively captured by tabular descriptors like the Orbital Field Matrix.

Despite the robustness of this framework, the research process revealed computational bottlenecks, particularly hardware constraints and the inherent limitations of standard DFT in resolving exact zero-gap boundaries for metallic states. To address these challenges and advance the framework, future work will focus on three primary dimensions:

1) **Architectural Refinement and Interpretability:** Implementing voxel-based 3D Convolutional Neural Networks (3D CNNs) to capture finer spatial charge density variations, alongside Graph Attention Networks (GATs) to identify specific atomic nodes and chemical bonds that drive electronic behaviors, thereby enhancing model interpretability.
2) **Multi-Fidelity Data Scaling:** Recalculating a representative subset of materials using the higher-fidelity HSE06 hybrid functional to correct systematic band gap underestimations inherent to the PBE functional. Concurrently, the dataset will be significantly scaled by integrating structural data from extensive open-source databases to enable multi-fidelity learning.
3) **Inverse Material Design:** Shifting the research paradigm from property prediction to target-driven generation. By deploying Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs), the framework will aim to inversely design novel crystal structures based on desired optoelectronic criteria, ultimately enabling the autonomous discovery of advanced functional materials.

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